

Assignment:- 4A

Binary to Gray code converter using logic gates

Binary → Gray code converter (implementation using logic gates)

1. Introduction

Binary code is the standard positional representation of numbers, whereas **Gray code** is a code in which two successive numbers differ in only one bit.

Converting binary to Gray code is widely used in digital systems, especially in encoders and error reduction.

2. Principle (General n-bit Conversion)

For an **n-bit** binary input:

- The **most significant Gray bit (G_{n-1})** is the same as the most significant binary bit (B_{n-1}).
 - $G_{n-1} = B_{n-1}$
 - Each remaining Gray bit is the **XOR** of the corresponding binary bit and the next higher binary bit:
 - $G_k = B_{k+1} \text{ XOR } B_k$ (for $k = n-2$ down to 0)
-

3. Boolean Equations (4-bit Example)

For a 4-bit binary input **B3 B2 B1 B0**, the Gray output bits are:

$$G_3 = B_3$$

$$G_2 = B_3 \text{ XOR } B_2$$

$$G_1 = B_2 \text{ XOR } B_1$$

$$G_0 = B_1 \text{ XOR } B_0$$

If only **AND, OR, NOT** gates are available, the XOR can be implemented as:

$$A \text{ XOR } B = (A \text{ AND NOT } B) \text{ OR } (\text{NOT } A \text{ AND } B)$$

4. Truth Table (4-bit Binary → Gray)

Binary Input (B3 B2 B1 B0)	Gray Output (G3 G2 G1 G0)
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

5. Implementation Notes

- **Inputs:** Binary number B3 B2 B1 B0
- **Outputs:** Gray code G3 G2 G1 G0
- **Logic circuit:**
 - G3 is directly connected to B3
 - G2 is XOR of B3 and B2
 - G1 is XOR of B2 and B1

- G0 is XOR of B1 and B0

Gate requirement for 4-bit converter:

- 3 XOR gates (or equivalent using AND/OR/NOT)
- 1 direct connection

6. Applications

- Digital communication (error minimization)
- Shaft encoders (to avoid spurious outputs)
- Karnaugh map simplification

